

DAY 07 – IPv4 CONFIGURATION ASSIGNMENT

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1. What is a network address?

A network address identifies a particular network or subnet. It is the first address in a subnet and is used to represent the network itself.

2. What is a broadcast address?

A broadcast address is the last address in a subnet. It is used to send data to all devices on that local network.

3. What is a usable host address?

A usable host address is an IP address that can normally be assigned to a device on a subnet. It is neither the network address nor the broadcast address.

4. What is the first usable IP address?

The first usable IP address is the address immediately after the network address. For example, in 192.168.1.0/24, the first usable IP is 192.168.1.1.

5. What is the last usable IP address?

The last usable IP address is the address immediately before the broadcast address. For 192.168.1.0/24, it is 192.168.1.254.

6. What does /24 mean?

/24 is CIDR notation. It means that 24 of the 32 IPv4 bits are used for the network portion and 8 bits are available for hosts.

7. What subnet mask corresponds to /24?

The subnet mask for /24 is 255.255.255.0.

8. What is a default gateway?

A default gateway is usually a router interface that forwards traffic from a local network to other networks, such as the Internet.

9. Why does a router interface need an IP address?

A router interface needs an IP address so it can communicate at Layer 3, identify the connected network, and route IP packets between networks.

10. What is a static IP configuration?

A static IP configuration is a manually assigned network configuration in which the IP address and related settings are intentionally fixed rather than automatically assigned by DHCP.

11. What does "show ip interface brief" display?

On Cisco devices, this command provides a quick summary of interfaces, including interface names, assigned IP addresses, and their status/protocol state.

12. What does "no shutdown" do conceptually on a Cisco interface?

The no shutdown command administratively enables a Cisco interface. It removes the shutdown state so the interface can become operational if the physical/link conditions are correct.

13. What does interface status "up/up" generally indicate?

It generally means the interface is administratively enabled and the line protocol is operational. In simple terms, the interface and its protocol are working.

14. Why should network configuration always be verified?

Verification helps confirm that the intended IP address, subnet, gateway, interface state, and other settings are correct. It can also reveal configuration or connectivity problems.

15. Why is IPv4 configuration important in cybersecurity?

Correct IPv4 configuration helps security professionals understand network boundaries, identify systems, monitor traffic, troubleshoot connectivity, and investigate suspicious network activity.

IPv4 EXERCISE – 192.168.100.0/24

Network Address: 192.168.100.0

First Usable IP: 192.168.100.1

Last Usable IP: 192.168.100.254

Broadcast Address: 192.168.100.255

PRACTICAL:-

- IPv4:- 10.83.207.182
- Subnet mask: 255.255.255.0
- Default gateway:- fe80::415:5dff:fe49:3r44%8
- DNS server:- 10.83.217.244

IPCONFIG /all

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Command Prompt
C:\Users\satya>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Satyam
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC #3
Physical Address. . . . . : 26-B2-B9-86-16-8B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC #5
Physical Address. . . . . : 2A-B2-B9-86-16-8B
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:
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PING 127.0.0.1

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Command Prompt
DHCP Server . . . . . : 10.83.207.244
DHCPv6 IAID . . . . . : 136622777
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-1C-E9-81-D4-AE-52-5B-90-3A
DNS Servers . . . . . : 10.83.207.244
                        2409:40d4:30dc:eb01::89
NetBIOS over Tcpi. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 24-B2-B9-86-16-8C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\satya>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```